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FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION ISLAMABAD

Version No.				
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ROLL NUMBER						

0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

COMPUTER SCIENCE
HSSC–I
SECTION – A (Marks 13)

Time allowed: 20 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصّہ اوّل لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Answer Sheet No. ____________

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پر کریں۔ **Invigilator Sign.** ____________

Fill the relevant bubble against each question according to curriculum: **Candidate Sign.** ____________

	Question	A	B	C	D	A	B	C	D
1.	Which gate produces a **HIGH** output when at least one input is **LOW**?	AND gate	NAND gate	OR gate	NOR gate				
2.	What is an efficient way to organize a collection of photos on your computer?	Naming them in random order	Grouping them by date or event	Deleting some photos randomly	Printing all photos and shuffling them				
3.	Which of the following represents a reliable source of information?	A peer-reviewed academic journal	A social media post without citation	A personal blog with no evidence	A random website with advertisements				
4.	What is the main objective of classification in Supervised learning?	To group data into clusters without labels	To predict continuous values	To reduce the dimensions of data	To assign predefined labels to input data				
5.	Which type of cloud deployment model is most suitable for organizations with strict regulatory requirements?	Private cloud	Public cloud	Hybrid cloud	Community cloud				
6.	Consider the list, **P= [2, 5, 3, 4]**, what will the statement **print (max(P))** return?	2	4	5	Error in statement				
7.	What is the primary goal of **abstraction** in computational thinking?	Writing detailed step–by–step instructions for solving a problem	Using a computer to solve mathematical equations	Repeating a process until a solution is found	Ignoring unnecessary details to focus on the important aspects of problem				
8.	Which type of feasibility study in SDLC involves evaluating whether the proposed system can be developed with available technology?	Legal feasibility	Schedule feasibility	Technical feasibility	Economic feasibility				
9.	Which of the following is an example of **open-ended** research question?	How many students use mobile phones in a class?	Does exercise reduce stress?	What are the effects of climate change on agriculture?	Is online education better than traditional education?				
10.	Which of the following can be classified as a **high-fidelity** prototype?	A detailed mobile app demo with clickable buttons	A hand-drawn wireframe of a website	A basic outline of an app on paper	A simple text document describing the app features				
11.	What is the main advantage of using a **simulation** instead of real-world testing?	Always gives perfect results	Risk-free and cost-effective	Eliminates the need for real-world experience	Requires no computer programming				
12.	What will be the result of $x=2 \mid 4$ Python statement?	0	2	4	6				
13.	Which of the following is an example of a service?	e-book	Laptop	Medical consultation	Pre-packaged food items				

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COMPUTER SCIENCE HSSC–I

Time allowed: 2:40 Hours **Total Marks Sections B and C: 62**

SECTION – B (Marks 42)

Q. 2 Answer the following parts briefly. **(14 x 3 = 42)**

<table>
<tr><td>(i)</td><td>Assume the variables x, y and z where x=4, y=10, and z=2. What value will be stored in result1, result2 and result3 after executing following Python statements?
a) result1 = y – x
b) result2 = y // z
c) result3 = x % z</td><td>03</td><td>OR</td><td>Trace the following pseudocode and complete the table below, which tracks the values of the variables at each iteration of the loop.
SET product =1
FOR i FROM 1 TO 3 DO
product = product * i
END FOR
PRINT product
<table><tr><td>Iteration No.</td><td>Value of i</td><td>Value of sum</td><td>Print</td></tr><tr><td>1</td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td></tr></table></td><td>03</td></tr>
<tr><td>(ii)</td><td>Write down any three features of data visualization.</td><td>03</td><td>OR</td><td>What is Permissioned Block chain network? Also list down its applications. (any two)</td><td>1+2</td></tr>
<tr><td>(iii)</td><td>Write down three comparisons between Symmetric and Asymmetric encryption.</td><td>03</td><td>OR</td><td>Differentiate between supervised and unsupervised learning with daily life examples.</td><td>03</td></tr>
<tr><td>(iv)</td><td>Why is prototyping important? Give three reasons.</td><td>03</td><td>OR</td><td>Write down three differences between Black Box and White Box testing methods.</td><td>03</td></tr>
<tr><td>(v)</td><td>Determine whether these variable names are valid or invalid. Provide reasons for any invalid variable names.
a) input b) your age c) marks1</td><td>03</td><td>OR</td><td>Why is two-factor authentication (2FA) important? Also give example.</td><td>2+1</td></tr>
<tr><td>(vi)</td><td>Briefly explain three harmful effects of increased connectivity on environment.</td><td>03</td><td>OR</td><td>Write down three positive impacts of AI systems.</td><td>03</td></tr>
<tr><td>(vii)</td><td>Which algorithm is more efficient: insertion sort or bubble sort? Justify your choice with two reasons.</td><td>1+2</td><td>OR</td><td>Correct any three errors in the following Python code:
temperature = 5;
if temperature > 30:
    print(“It's hot)
elif temperature = 5:
    display(“It's cold”)
else:
print(“The weather is mild”)</td><td>03</td></tr>
<tr><td>(viii)</td><td>Evaluate the following expression in Python using the proper order of operations:
2+2**3//2*4%3-1</td><td>03</td><td>OR</td><td>Compare Bus and Mesh network topologies with respect to design, scalability and reliability.</td><td>03</td></tr>
<tr><td>(ix)</td><td>Briefly explain any three guidelines to conduct a better qualitative interview.</td><td>03</td><td>OR</td><td>Write down any three key steps in developing the Minimum Viable Product.</td><td>03</td></tr>
<tr><td>(x)</td><td>Write a Python program that asks the user to input name and marks. Also displays the entered name and marks.</td><td>03</td><td>OR</td><td>Draw truth table for the Boolean function.
$F=\overline{X}YZ+X\overline{Y}Z$</td><td>03</td></tr>
<tr><td>(xi)</td><td>What is the importance of Digital literacy? Give three reasons.</td><td>03</td><td>OR</td><td>Write down any three tasks suitable for computing devices.</td><td>03</td></tr>
<tr><td>(xii)</td><td>Why is horizontal scalability considered more suitable for cloud-based applications? Give three reasons.</td><td>03</td><td>OR</td><td>What will be generated by the following Python code?
country=[“Pakistan”, “Iran”, “China”]
country [2]= “Turkey”
country.append(“Oman”)
print(country)</td><td>03</td></tr>
<tr><td>(xiii)</td><td>Compare Parameter and Statistics with daily life example.</td><td>03</td><td>OR</td><td>Write Python statements considering turtle graphics to perform the following actions:
a) Change the pen colour to blue.
b) Move the turtle backward by 150 pixels.
c) Rotate the turtle 90 degrees to the left.</td><td>03</td></tr>
<tr><td>(xiv)</td><td>What is the deployment phase in SDLC? How does the pilot deployment method differ from the parallel deployment method?</td><td>1+2</td><td>OR</td><td>What is a sample? Why is it often used instead of studying an entire population?</td><td>1+2</td></tr>
</table>

SECTION – C (Marks 20)

Attempt the following questions. **(4 x 5 = 20)**

<table>
<tr><td>Q.3</td><td>Write down any five differences between Waterfall and Agile software development models.</td><td>05</td><td>OR</td><td>Explain the differences between Correlation and Causation with examples.</td><td>05</td></tr>
<tr><td>Q.4</td><td>Simplify the Boolean Function F, using Karnaugh Map. Aso draw the logic circuit for simplified expression.
$F=\overline{A}\overline{B}C+\overline{A}BC+A\overline{B}\overline{C}+A\overline{B}C+AB\overline{C}+ABC$</td><td>05</td><td>OR</td><td>Write a Python program that reads two numbers and prints the minimum number by using function.</td><td>05</td></tr>
<tr><td>Q.5</td><td>Use the Linear Search algorithm to find the number 53 from the given set of numbers:
<table><tr><td>89</td><td>50</td><td>36</td><td>42</td><td>11</td><td>53</td><td>65</td><td>72</td><td>24</td></tr></table></td><td>05</td><td>OR</td><td>Write a Python program that inputs a number and prints its multiplication table from 1 to 10 using for loop.</td><td>05</td></tr>
<tr><td>Q.6</td><td>What is meant by Internet of Things (IoT)? Describe any four applications of IoT technology.</td><td>1+4</td><td>OR</td><td>What are information sources? Explain any four categories of information sources.</td><td>1+4</td></tr>
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